

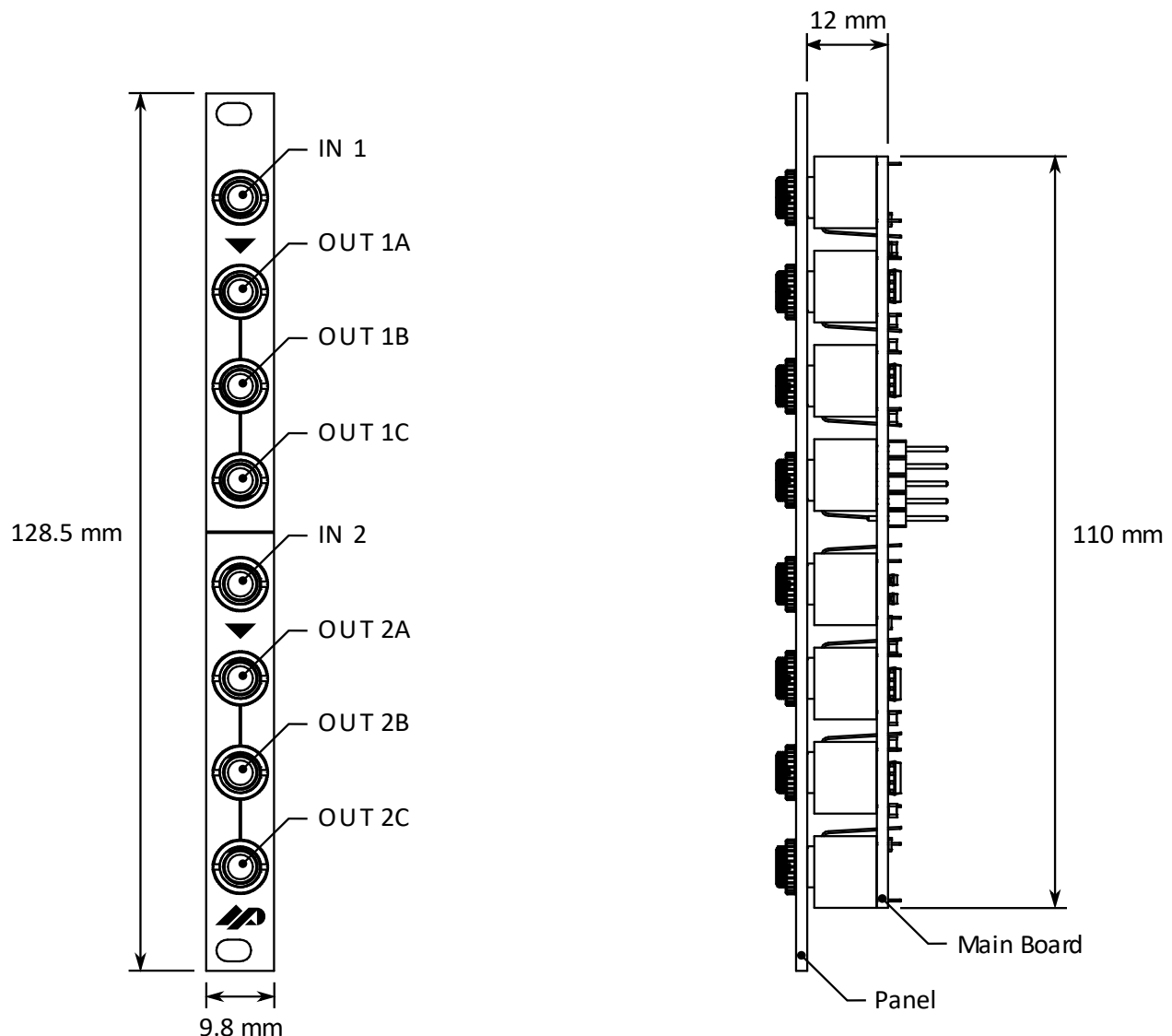
Description

The μMult is a dual buffered multiple in a compact format. Signals at the inputs will be replicated and buffered to the outputs without affecting any upstream circuitry. This module features two operational modes that allow either one or two inputs while utilizing all available outputs.

Features

- Compact 2hp, 12mm-deep form factor
- Switchable functionalities
 - Single 1 ▶ 6 buffered multiple
 - Dual 1 ▶ 3 buffered multiples
- Separate buffer circuits eliminate crosstalk between inputs
- Power input reverse-polarity protection
- Current Draw: +12V: 16mA, -12V: 16mA

Module Layout



Functionality

Single-Mult Mode

When an input is connected to *only* the **IN 1** jack, the module functions as a single 1►6 buffered multiple. Signals present at **IN 1** will be replicated to outputs **OUT 1A – OUT 1C** as well as outputs **OUT 2A – OUT 2C**.

Dual-Mult Mode

When an input is connected to *both* the **IN 1** and **IN 2** jacks, the module functions as dual 1►3 buffered multiples. Signals present at **IN 1** will be replicated to outputs **OUT 1A – OUT 1C**, and signals present at **IN 2** will be replicated to outputs **OUT 2A – OUT 2C** separately.

Bill of Materials

Value	Qty.	Name on Board	Description	Footprint
100k	2	R1,R5	100k 0.125W Resistor	0805
1k	6	R2,R3,R4,R6,R7,R8	1k 0.125W Resistor	0805
100n	8	C1,C2,C3,C4,C5,C6,C7,C8	100n 50V Capacitor	0805
TL072	4	U1,U2,U3,U4	TL072 Dual Op Amp	SOIC-8 3.9mm
Schottky Diode	2	D1,D2	Schottky Diode 50V	SOD-123FL
3.5mm Jack	8	J1,J2,J3,J4,J5,J6,J7,J8	3.5mm Mono Switched Jack	PJ-3001F
2.54mm Header	1	J9	2.54mm Header 5x1	2.54mm 5-Pin Vertical

Schematic

